

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2025

EC1001-1 – BASIC ELECTRONICS

(For AC, EC, EE, VL)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The maximum efficiency of a full wave rectifier is
A) 38.2% B) 40.6%
C) 81.2% D) 100%
2. An input signal applied to a rectifier circuit is $23 \cos 314t$. The value of V_m is
A) 46 V B) 23 V
C) 32.53 V D) 50 V
3. Which one is correct for a full wave bridge rectifier circuit?
A) $I_m = V_m/(R_L)$ B) $I_m = V_m/(R_f + R_s + R_L)$
C) $I_m = V_m/(R_s + R_L)$ D) $I_m = V_m/(2R_f + R_s + R_L)$
4. A 24 volt, 600 mW Zener diode is used in a voltage regulator circuit. The Zener current is
A) 10 mA B) 32 mA
C) 25 mA D) 300 mA
5. For a BJT $I_c = 1$ mA and $I_B = 25$ micro A. The value of β is
A) 0.04 B) 0.4
C) 4 D) 40
6. For a BJT $I_c = 1$ mA and $I_B = 25$ micro A. The value of I_E is
A) 10.25 mA B) 1.25 mA
C) 1.025 mA D) 125 micro A
7. The operating point of the transistor when operated as a switch is
A) Cut off and Saturation B) Cut off and Active
C) Active D) Active and Saturation
8. To use the transistor as an amplifier
A) EB-FB, CB-FB B) EB-RB, CB-FB
C) EB-FB, CB-RB D) EB-RB, CB-RB
9. There are _____ depletion regions in a transistor
A) 1 B) 2
C) 3 D) 4
10. The drain current in an FET is controlled by
A) Gate voltage B) Drain voltage
C) Collector voltage D) Source voltage
11. The voltage gain of an OPAMP voltage follower circuit is
A) Infinity B) Zero
C) One D) 100
12. The process of modulation is mainly used to
A) Reduce antenna height B) Allow multiplexing
C) Reduce interference D) All the above

13. The input and output of an inverting amplifier are related as
 A) Output signal is in phase w.r.t the input B) Output signal is 90° shifted w.r.t. the input
 C) Output signal is out of phase w.r.t the input D) Output signal is -90° shifted w.r.t. the input
14. The input impedance of an ideal OPAMP is
 A) Negative B) Medium
 C) High D) Zero
15. An ideal OPAMP will have
 A) Narrow bandwidth B) Infinite bandwidth
 C) Zero bandwidth D) Medium bandwidth
16. The main advantage of the negative feedback system is
 A) Controlled gain B) Increased bandwidth
 C) Good stability D) All the above
17. An embedded system communicates with the outside world through
 A) Memory B) Input
 C) Output D) Peripherals
18. The brain of an embedded system is
 A) Sensors B) Microcontroller
 C) Power supply D) Actuator
19. The best example for a wireless channel is
 A) Fiber optic cable B) Coaxial cable
 C) Twisted pair cable D) Free space
20. Best example for atmospheric noise is
 A) Rain B) Traffic noise
 C) Industrial noise D) Sound pollution

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the VI characteristics of a npn transistor connected in common emitter configuration with necessary diagrams.	6	L*2	1	1
	b) A circuit consists of a silicon diode in series with a $2.7\text{ k}\Omega$ resistor and a battery. Draw the circuit diagram and calculate the battery voltage if the forward current is 1.96 mA .	4	L3	1	1
	c) In a bridge rectifier, a diode whose internal resistance is $20\ \Omega$ is used to supply power to a $1\text{ k}\Omega$ load from 110 V(rms) supply. Calculate the (i) DC load current (ii) DC voltage (iii) DC power delivered to the load.	6	L3	1	1
2.	a) Explain the working of transistor as a switch with neat circuit diagram.	4	L2	1	1
	b) With neat diagrams explain the construction, operation and drain characteristics of a n-channel enhancement type MOSFET.	8	L2	2	1
	c) If the base current in a transistor is $20\ \mu\text{A}$ when the emitter current is 6.4 mA , Calculate the values of α and β . Also calculate the collector current.	4	L3	1	1
3.	a) Explain the construction and working of the Half wave rectifier circuit with necessary diagrams and waveforms. Derive the expression for I_{dc} , V_{dc} and P_{dc} .	10	L2	1	1
	b) A Zener diode voltage regulator has supply voltage $V_s = 20\text{ volts}$, the resistance in series with the dc supply is $200\ \Omega$ and the load resistance $R_L = 1\text{ k}\Omega$, reverse breakdown voltage of Zener is, $V_z = 10\text{ V}$. Calculate the current I_s from the dc supply and current through the Zener diode, I_z . Draw the circuit diagram for following specifications.	6	L3	2	1

Unit – II

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|-------|--|----|----|---|---|
| 4. a) | Prove that the closed loop voltage gain of the voltage series negative feedback amplifier is $A_f = \frac{A}{1+A\beta}$. Draw the necessary diagram. | 6 | L2 | 4 | 1 |
| b) | Explain the Hartley oscillator circuit using neat diagrams and necessary equations. | 6 | L2 | 4 | 1 |
| c) | Explain the Voltage follower circuit using Op-amp with neat circuit diagram and necessary waveforms. | 4 | L2 | 3 | 1 |
| 5. a) | Explain the following circuits designed using OP-AMP:
(i) Differentiator
(ii) Inverting amplifier | 10 | L2 | 3 | 1 |
| b) | In an oscillator, $C_1 = C_2 = C$ and $L = 100 \mu\text{H}$. The frequency of oscillations is 500 kHz. Calculate the value of C and also draw the circuit diagram. | 6 | L3 | 4 | 1 |
| 6. a) | Explain the operation of Astable multivibrator constructed using IC 555 Timer with necessary diagrams and waveforms. | 6 | L2 | 4 | 1 |
| b) | An Astable Multivibrator using IC555 times generates a square wave of 4 kHz and duty cycle of 50%. Calculate the value of R_A and R_B required. Assume the capacitor of $0.1 \mu\text{F}$ is used. | 6 | L3 | 4 | 1 |
| c) | In a RC phased shift oscillator, the phase shift network uses resistors of $4.7 \text{ k}\Omega$ and the capacitors of $0.47 \mu\text{F}$. Determine the frequency of oscillations. | 4 | L3 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Explain the elements of embedded systems with neat diagrams. | 6 | L2 | 5 | 1 |
| b) | Compare the microprocessor and microcontroller. | 4 | L2 | 5 | 1 |
| c) | What is modulation? why do you think the message signal is to be modulated before the transmission? Explain. | 6 | L2 | 5 | 1 |
| 8. a) | Explain the following with reference to wireless communication systems: (i) frequency reuse (ii) Hand off | 6 | L2 | 5 | 1 |
| b) | Discuss the different types of channels used in a communication system with suitable examples. | 6 | L2 | 5 | 1 |
| c) | Briefly explain the role of sensors and actuators in an embedded system. | 4 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
